

Tugas code read image, show image, filter color image, filter color video

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Kode yang Digunakan

```
import cv2
import numpy as np

# READ IMAGE

image = cv2.imread("rb.png")

if image is None:
    print("Gambar tidak ditemukan!")
    exit()

# FILTER COLOR IMAGE

# Grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Filter Green
green = image.copy()

green[:, :, 0] = 0
green[:, :, 2] = 0

# Filter Pink
pink = image.copy()

pink[:, :, 1] = 0

# SHOW IMAGE

cv2.imshow("Gambar Asli", image)
cv2.imshow("Grayscale", gray)
cv2.imshow("Green", green)
cv2.imshow("Pink", pink)
```

```

# PERUBAHAN WARNA MENGGUNAKAN HSV

hue = 0

while True:

    hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)

    hsv[:, :, 0] = hue

    colorful = cv2.cvtColor(hsv, cv2.COLOR_HSV2BGR)

    cv2.imshow("Colorful", colorful)

    hue += 1

    if hue >= 180:
        hue = 0

    if cv2.waitKey(30) == 27:
        break

cv2.destroyAllWindows()

# FILTER COLOR VIDEO – WEBCAM

cap = cv2.VideoCapture(0)

while True:

    ret, frame = cap.read()

    if not ret:
        print("Kamera tidak terbaca")
        break

    green = frame.copy()

    # Mengurangi channel Blue
    green[:, :, 0] = green[:, :, 0] * 0.3

    # Mempertahankan channel Green
    green[:, :, 1] = np.minimum(
        green[:, :, 1] * 1,
        255
    )

    # Mengurangi channel Red
    green[:, :, 2] = green[:, :, 2] * 0.3

    cv2.imshow("Kamera Asli", frame)
    cv2.imshow("Filter Hijau", green)

    # Tekan Q untuk keluar
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break

```